Expt. No. 4

Title: To design and verify the layout of CMOS inverter.

Rubric	*Score (1 to 5)*
Timely Submission	
Result Interpretation and Output	
Program Logic and Concept Understanding	
Total	

Performed On: _____

Sign:_____

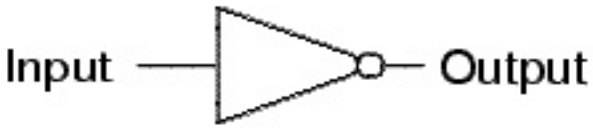
Experiment No. 4

Aim: To design and verify the layout of CMOS inverter.

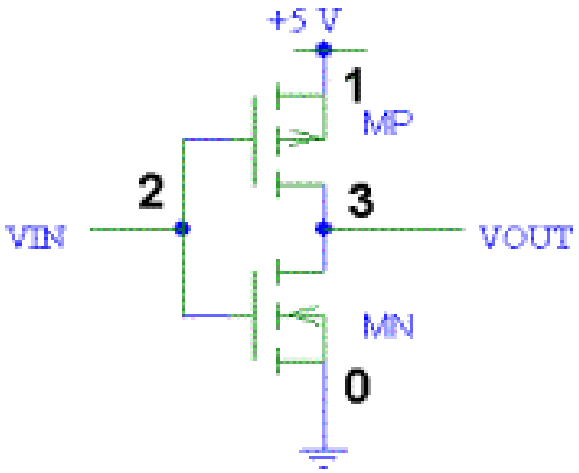
Software Required: Micro-wind software

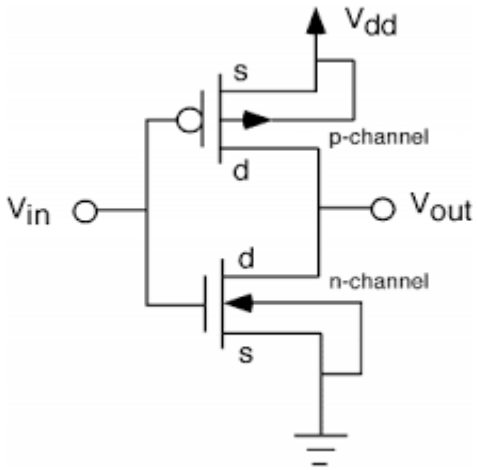
Hardware: computer

Circuit Diagram:



Input	Output
1	0
0	1





Theory:

The inverter is universally accepted as the most basic logic gate doing a Boolean operation on a single input variable. Fig.1 depicts the symbol, truth table and a general structure of a CMOS inverter. As shown, the simple structure consists of a combination of an pMOS transistor at the top and a nMOS transistor at the bottom.

CMOS is also sometimes referred to as **complementary-symmetry metal–oxide–semiconductor**. The words "complementary-symmetry" refer to the fact that the typical digital design style with CMOS uses complementary and symmetrical pairs of p-type and n-type metal oxide semiconductor field effect transistors (MOSFETs) for logic functions. Two important characteristics of CMOS devices are high noise immunity and low static power consumption. Significant power is only drawn while the transistors in the

CMOS device are switching between on and off states. Consequently, CMOS devices do not produce as much waste heat as other forms of logic, for example transistor-transistor logic (TTL) or NMOS logic, which uses all n-channel devices without p-channel devices. Some important condition to be kept in doing layout are

1] CMOS logic is based on a pair of NMOS and PMOS transistor, two different transistors have to be in layer, so that it will be a good for reading software of both transistors.
2] There must be a single nail for VDD as well as VDS.
3] Width of PMOS to be kept at a value of double for as compared as to NMOS.
4] Also follow the layout rules, it keeps checking DRC at every step so as to a void error in output at the end of simulator.

Procedure:

1] Start micro-wind software for layout.

2] Draw the CMOS inverter following the design.

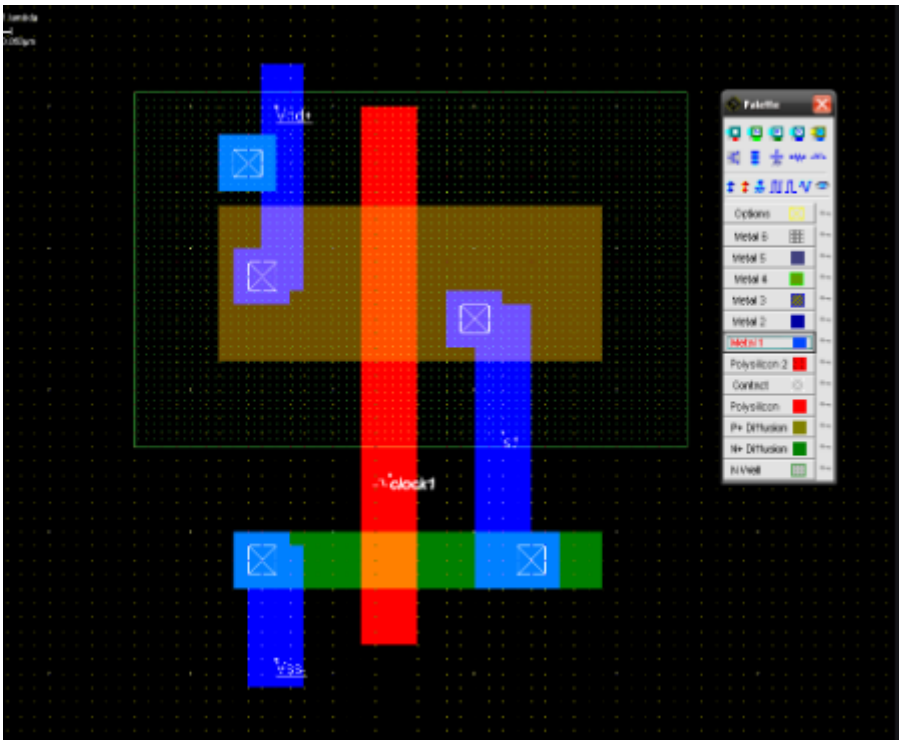
3] Click on DRC at every stage of design to avoid error.

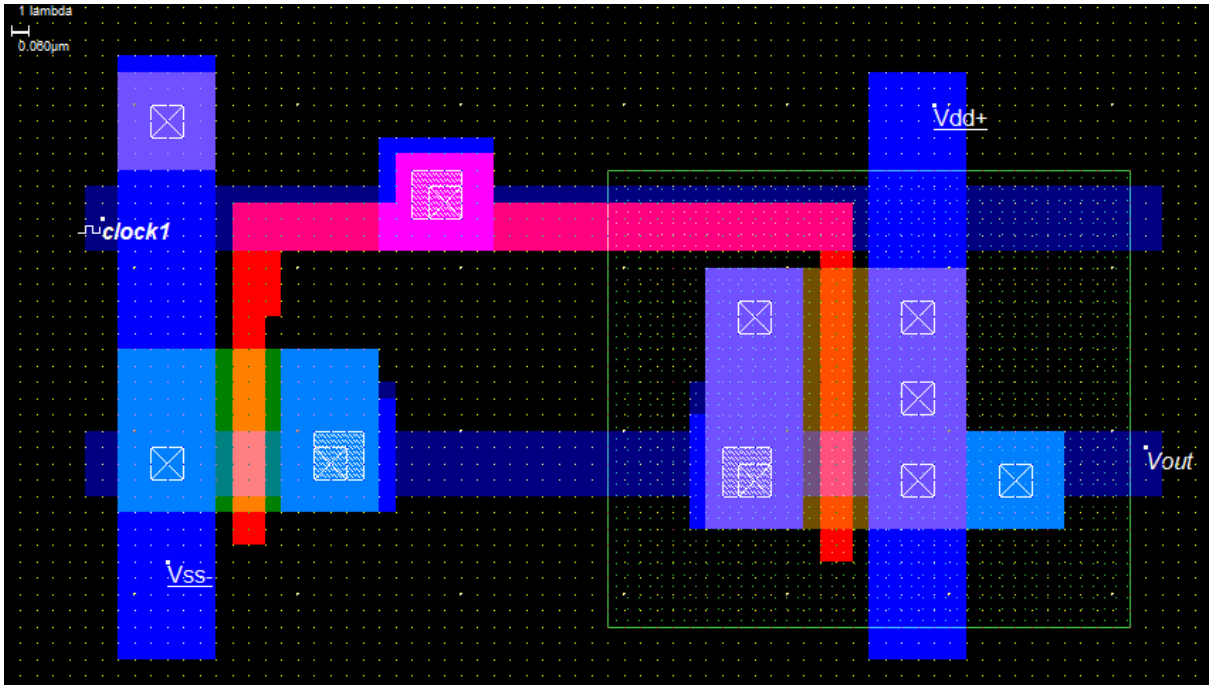
4] Observe various output while using simulation output.

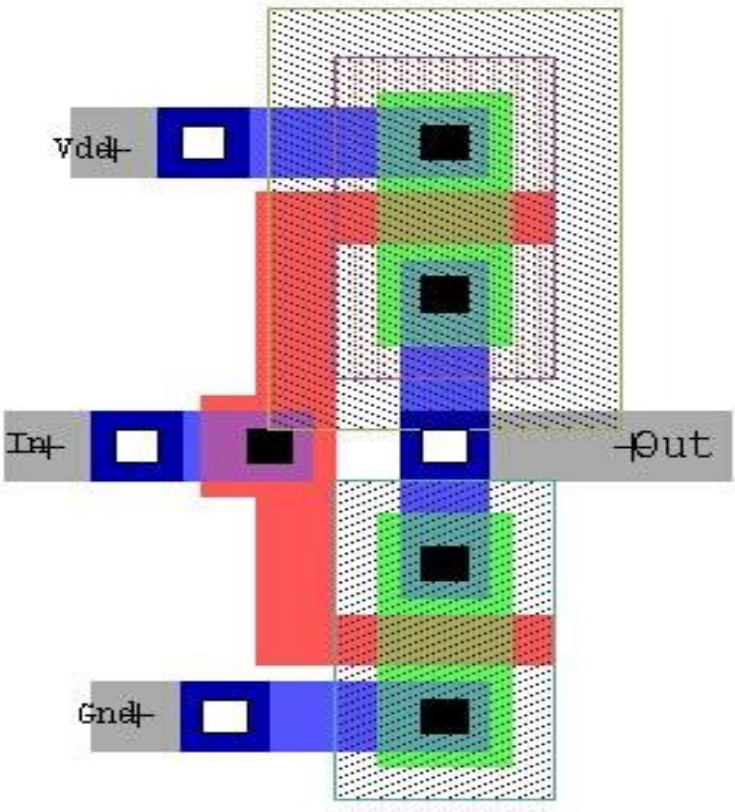
5] Software design and finally run a DRC.

6] Record the output waveform observed

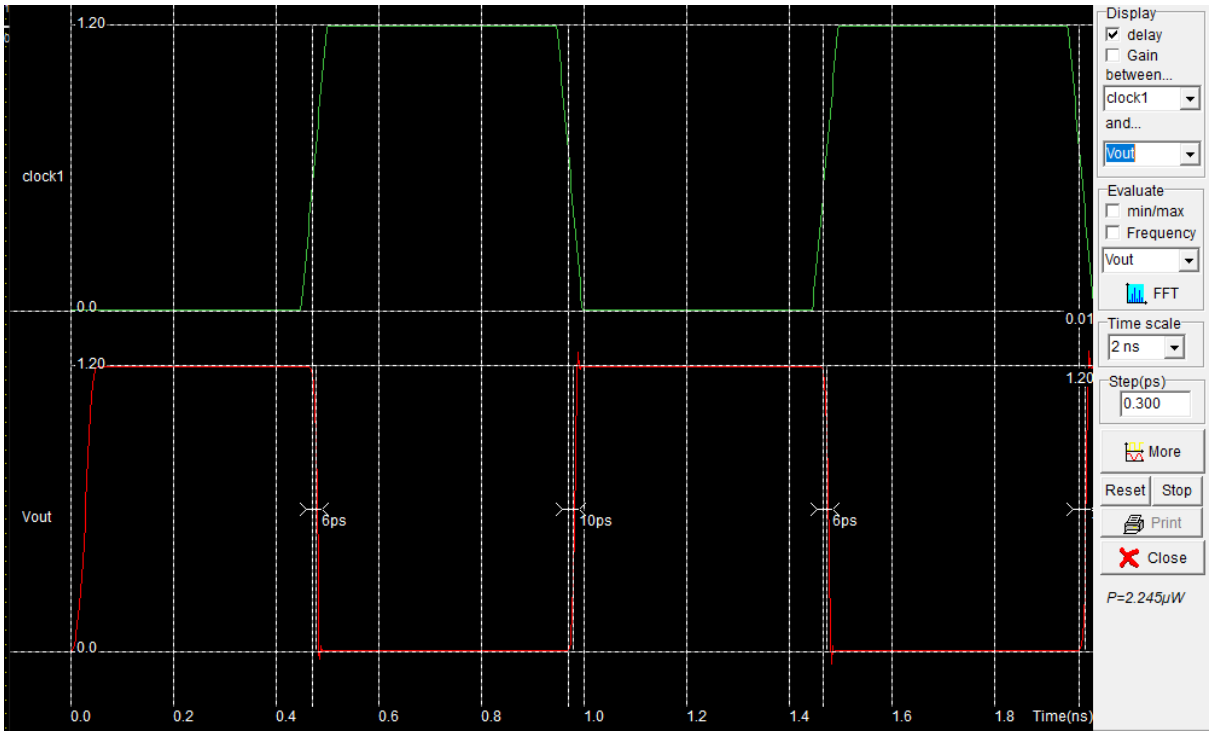
Layout:

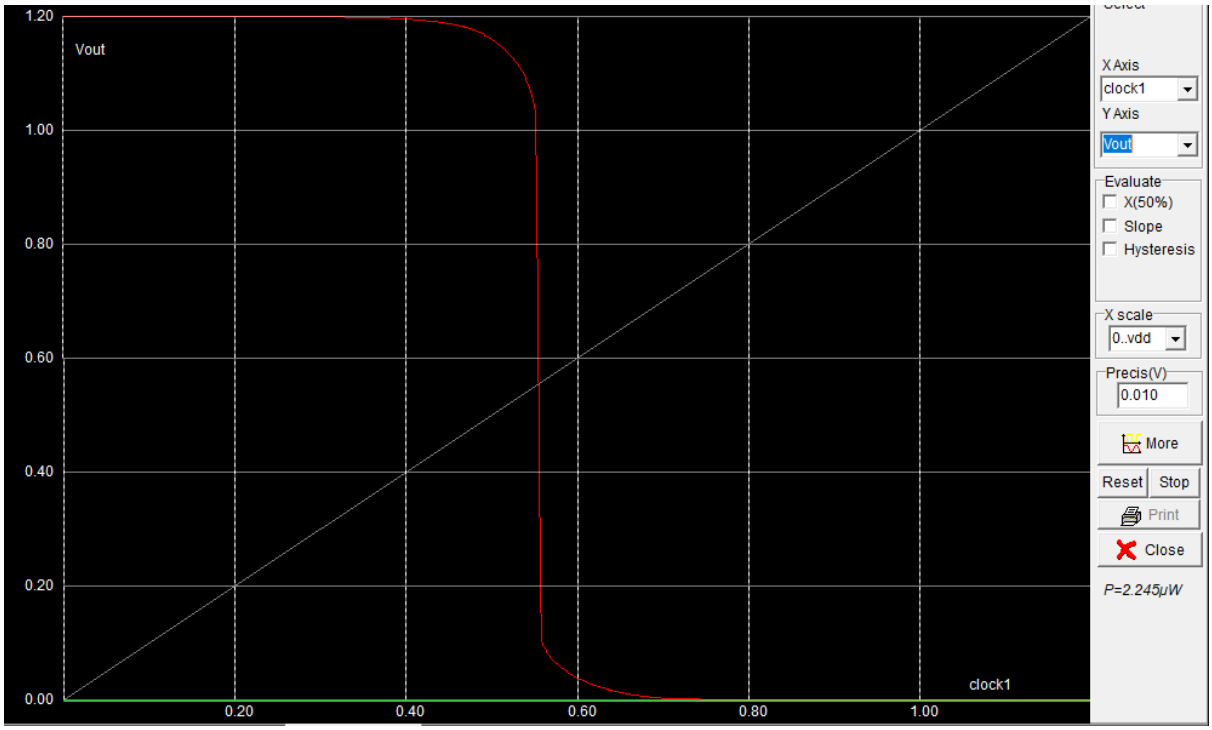


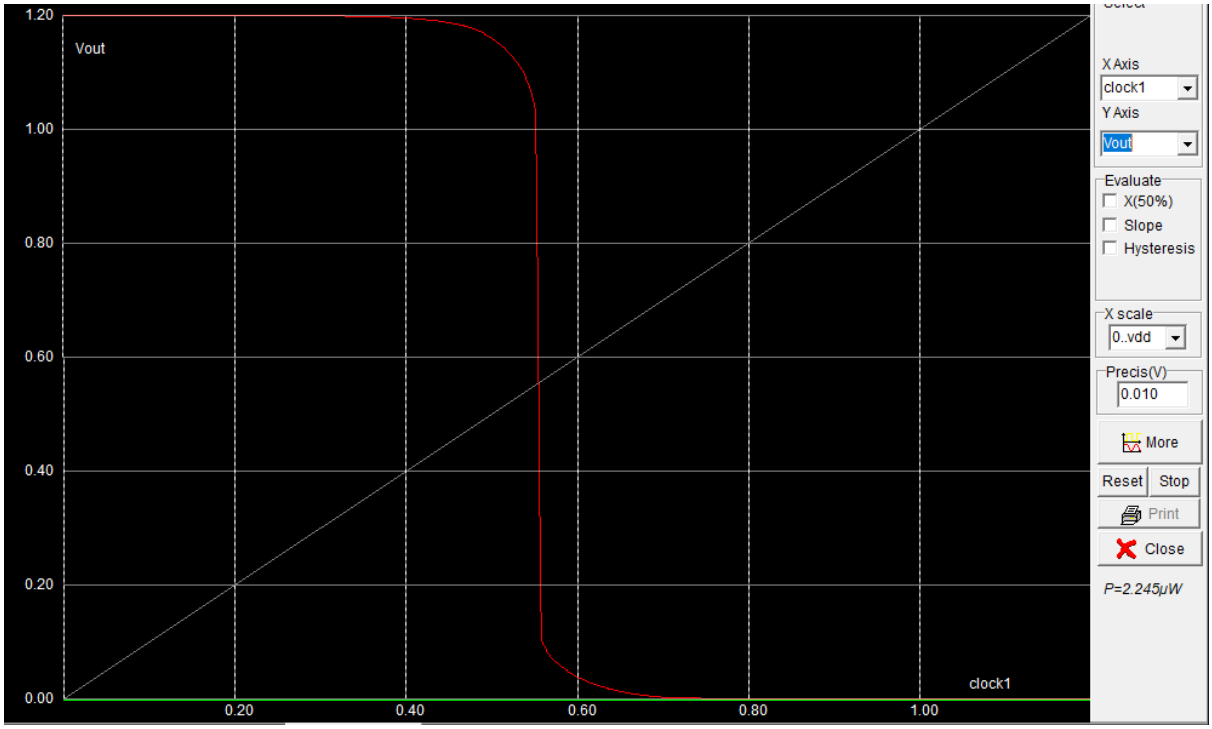

1 lambda
0.060µm
Vdd+
clock1
Vout
Vss-

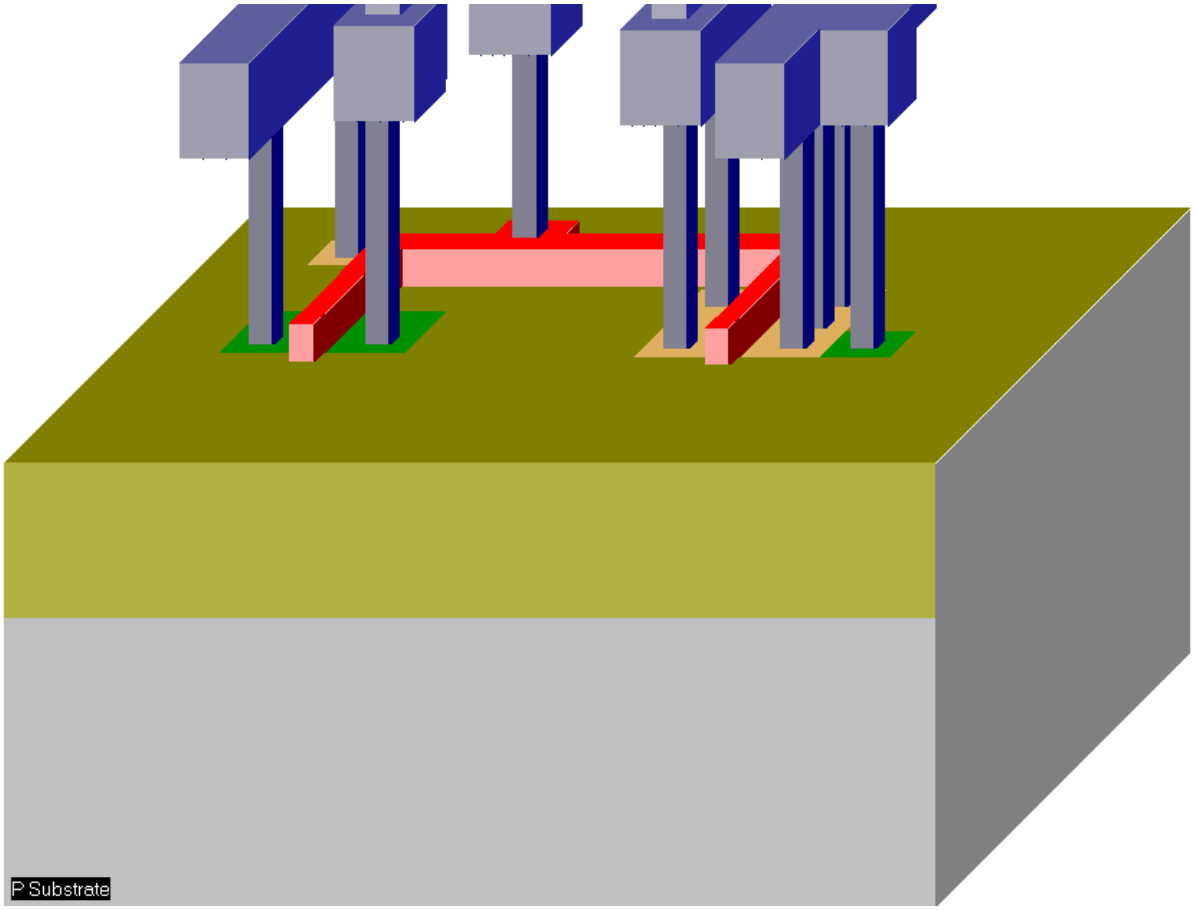

Vdd
In
Out
Gnd

Output:


1.20
clock1
0.0
1.20
Vout
0.0
6ps
10ps
6ps
0.01
1.20
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
1.8
Time(ns)
Display
delay
Gain
between...
clock1
and...
Vout
Evaluate
min/max
Frequency
Vout
FFT
Time scale
2 ns
Step(ps)
0.300
More
Reset
Stop
Print
Close
P=2.245µW


1.20
Vout
1.00
0.80
0.60
0.40
0.20
0.00
0.20
0.40
0.60
0.80
1.00
clock1
X Axis
clock1
Y Axis
Vout
Evaluate
X(50%)
Slope
Hysteresis
X scale
0..vdd
Precis(V)
0.010
More
Reset
Stop
Print
Close
P=2.245µW


1.20
1.00
0.80
0.60
0.40
0.20
0.00
Vout
0.20
0.40
0.60
0.80
1.00
clock1
X Axis
clock1
Y Axis
Vout
Evaluate
X(50%)
Slope
Hysteresis
X scale
0..vdd
Precis(V)
0.010
More
Reset
Stop
Print
Close
P=2.245µW


P Substrate

Pre Lab Questions:

1 For Cmos Logic, Give The Various Techniques You Know To Minimize Power Consumption?
2 Why Do We Gradually Increase The Size Of Inverters In Buffer Design? Why Not Give The Output Of A Circuit To One Large Inverter**?**

Post Lab Questions:

1 Why Is The Substrate In Nmos Connected To Ground And In Pmos To Vdd?
2 What Is The Fundamental Difference Between A Mosfet And Bjt?

Conclusion: Thus, we have designed and verified the layout of CMOS inverter circuit using micro-wind waveform.

CO's Covered:

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